

Power Query in Power BI

Hands-On Data Transformation Exercise

Level: Beginner to Intermediate **Duration:** 60–90 minutes **File:** PowerQuery_Exercise_Dataset.xlsx

1. Overview

This exercise uses a deliberately messy retail sales dataset. Your job is to use the Power Query Editor in Power BI Desktop to clean, shape, and combine the data until it is analysis-ready. Every task lists the goal, the clicks, and the expected result so you can check your own work.

The dataset has three tables (sheets):

Sheet	What it is	Notes
Sales	24 sales transactions (the fact table)	Contains the mess: text dates, untidy names, inconsistent regions, blanks, and a duplicate row
Products	6-row product lookup table	Clean — used to enrich Sales via a Merge
RegionSalesWide	Region totals by month (wide layout)	Used for the Unpivot bonus task

Data-quality issues planted in the Sales table:

- OrderDate is stored as text, not a real date.
- CustomerName has leading/trailing spaces and inconsistent capitalisation.
- Region is inconsistent — e.g. "north", "North ", and "NORTH" all mean the same place.
- Quantity has two values stored as text and one blank cell.
- UnitPrice has one blank cell.
- One transaction (ORD-1020) is fully duplicated.

2. Setup — Load the Data

1. Open **Power BI Desktop**.
2. On the Home ribbon, click **Get Data** → **Excel workbook**, then browse to the exercise file.
3. In the Navigator, tick **Sales**, **Products**, and **RegionSalesWide**.
4. Click **Transform Data** (not Load) to open the Power Query Editor.

Expected result: The Power Query Editor opens with three queries listed on the left.

3. Task 1 — Fix Data Types

Select the **Sales** query.

5. Click the icon on the **OrderDate** column header → choose **Date**.
6. Set **Quantity** to **Whole Number** and **UnitPrice** to **Decimal Number**.
7. If a type-change prompt appears, choose **Replace current**.

Expected result: OrderDate shows a calendar icon; the two text quantities convert to numbers.

4. Task 2 — Clean the Customer Names

8. Select the **CustomerName** column.
9. Transform tab → **Format** → **Trim** (removes extra spaces).
10. Transform tab → **Format** → **Clean** (removes non-printing characters).
11. Transform tab → **Format** → **Capitalize Each Word**.

Expected result: Names read consistently, e.g. " ahmad ismail" becomes "Ahmad Ismail".

5. Task 3 — Standardise the Region Column

12. Select the **Region** column.
13. Apply **Format** → **Trim**, then **Format** → **Capitalize Each Word**.
14. Click the filter arrow on Region to confirm only four distinct values remain.

Expected result: Region contains exactly four clean values: North, South, East, West.

6. Task 4 — Handle Blanks and Remove Duplicates

15. Select **Quantity** and **UnitPrice** (Ctrl-click both).

16. Transform tab → **Replace Values**: leave "Value to find" empty is not allowed, so instead right-click → **Replace Values** and replace **null** with **0**. (Tip: replacing null works from the column's right-click menu.)

17. Select the **OrderID** column → Home tab → **Remove Rows** → **Remove Duplicates**.

Expected result: The two blank cells become 0, and the row count drops from 24 to 23 (duplicate ORD-1020 removed).

7. Task 5 — Add Calculated Columns

18. Add Column tab → **Custom Column**. Name it **TotalSales** with the formula:

= [Quantity] * [UnitPrice]

19. Add Column tab → **Conditional Column**. Name it **SalesBand**: if TotalSales is greater than or equal to **1000** then "High", else "Standard".

20. Set TotalSales data type to **Decimal Number**.

Expected result: Two new columns: TotalSales (a number) and SalesBand (High / Standard).

8. Task 6 — Merge with the Products Table

21. With **Sales** selected, Home tab → **Merge Queries**.

22. Choose **Products** as the second table.

23. Click the **ProductCode** column in both tables to link them; keep join kind **Left Outer**.

24. Click the expand icon on the new Products column and bring in **ProductName** and **Category** (untick "Use original column name as prefix").

Expected result: Each sales row now shows its ProductName and Category.

9. Bonus Task — Unpivot the Monthly Table

25. Select the **RegionSalesWide** query.

26. Select the **Jan**, **Feb**, and **Mar** columns.

27. Transform tab → **Unpivot Columns**.

28. Rename **Attribute** to **Month** and **Value** to **Amount**.

Expected result: The 4-row wide table becomes a tidy 12-row table: Region, Month, Amount.

10. Finish — Close & Apply

29. Home tab → **Close & Apply**.

30. Review the **Applied Steps** pane for each query — every transformation you made is recorded and repeatable.

Expected result: Data loads into the model, clean and ready for reporting.

11. Self-Check Summary

Check	Correct outcome
Sales row count	23 rows (was 24)
OrderDate type	Date
Distinct regions	4 (North, South, East, West)
Blank Quantity / UnitPrice	Replaced with 0
New columns in Sales	TotalSales, SalesBand, ProductName, Category
RegionSalesWide (unpivoted)	12 rows: Region, Month, Amount

Discussion prompt: Which single step would break if next month's file used a different date format? How does keeping steps in the Applied Steps pane make the whole process reusable?